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Subject: Manuscript submission to PRX Life.

Dear Editor,

Please find attached our manuscript titled “Algebraic and diagrammatic methods for the rule-based modeling of multi-particle complexes”, which we kindly ask to be considered for publication in PRX Life.

The formation, dissolution, and dynamics of multi-particle complexes occurs in many different stochastic chemical systems. This is especially so in biophysics (our native discipline), where gene expression is controlled in diverse ways by the assembly of large protein-nucleic acid complexes.

In 1976, Masao Doi introduced field theory methods for modeling stochastic chemical systems. As in quantum field theory, Doi’s approach uses an operator algebra to model the creation and annihilation of classical particles. This approach has found a variety of applications, particularly in the study of diffusion-limited reactions. Doi’s approach, however, is unable to model the assembly of pre-existing particles into multi-particle complexes.

Starting in the 2000’s, multiple groups developed software packages for computationally simulating chemical systems involving large multi-particle complexes. Given a specified set of reaction rules, these software tools simulate the dynamic growth and dissolution of all multi-particle complexes that can result. The rule-based nature of these methods is key, as a small number of rules can lead to a very large (or even infinite) number of possible complexes. The formal underpinnings of these algorithms, however, do not facilitate analytic computations using the established tools of equilibrium and nonequilibrium statistical mechanics.

Here we bridge these two lines of research by developing rule-based algebraic methods for studying stochastic chemical systems of multi-particle complexes. We also develop formal diagrammatic methods that aid in the use of this algebra. As shown in our paper, our mathematical methods can be used to analytically study systems both in and out of thermal equilibrium. We further develop a stochastic simulation algorithm based on our formalism. We are therefore hopeful that this work will open many previously intractable systems to theoretical study, as well as enable the development of new computational approaches for studying these systems.

We believe this work will appeal to a diverse readership, including theoretical physicists studying stochastic chemical systems, computational physicists in the field of rule-based modeling, and to biophysicists who are interested in quantitatively modeling complex molecular systems more generally. For these reasons we believe our manuscript is well-suited for publication in PRX Life.

We suggest the following experts as potential reviewers:

- Jeremy Gunawardena, Harvard Medical School, jeremy_gunawardena@hms.harvard.edu
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Thank you for your consideration,

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